Introduction to Probability and Statistics
(EE 354 / CE 361)

Sets, Sample Space, Probabilistic Models, Axioms

Today's Agenda

- Recall: Probability is a mathematical framework for dealing with uncertainty
- Probabilistic Models
  - Sample Space
  - Probability Law
- Axioms of Probability
- Simple Examples

Some Basic Set Notation

Sets

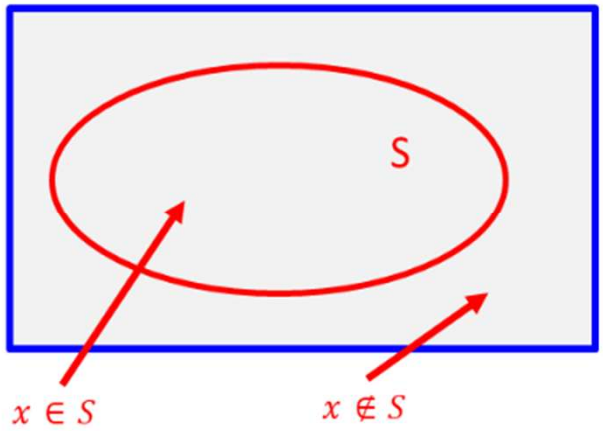
- A Collection of distinct elements

$\{a, b, c, d\}$ Finite

R: *real numbers* Infinite

$\{x \in R: \cos(x) > \frac{1}{2}\}$

Ω: universal set

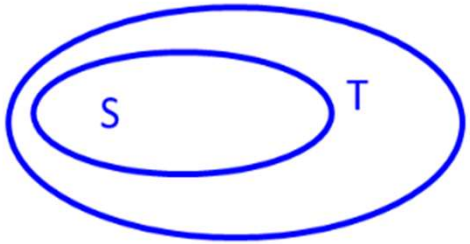


S^c: $x \in S^c$ *if* $x \in \Omega, x \notin S$

$(S^c)^c = S$

$\Omega^c = \phi$ ϕ: *empty set*

$S \subset T$:



Example:

- **Question**: The amount of rain in Karachi in the month of August is X cm.
- Question No 3b: What is X? Discuss
- Question No 3c: Predict any formula for X
- Question No 3d: Discuss the variability in X from 2000 to 2017
- Question No 3e: Draw or predict a histogram of X?

Some Basic Set Notation

$\{\,\}$

$\{H, T\}$

$\emptyset = \{\,\}$

$S = \{x_1, x_2, x_3, x_4, \ldots\}$

$\{x \mid x\ satisfies\ P\}$

$S^C = \{x \in \Omega \mid x \notin S\}$

Some Basic Set Notation

$$S \cup T = \{x \mid x \in S \text{ or } x \in T\}$$

$$S \cap T = \{x \mid x \in S \text{ and } x \in T\}$$

$$\bigcup_{n=1}^{\infty} S_n = S_1 \cup S_2 \cup \cdots = \{x \mid x \in S_n \text{ for some } n\}$$

$$\bigcap_{n=1}^{\infty} S_n = S_1 \cap S_2 \cap \cdots = \{x \mid x \in S_n \text{ for all } n\}$$

Set Algebra

$$S \cup T = T \cup S$$
$$S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$$
$$(S^C)^C = S$$
$$S \cup \Omega = \Omega$$

$$S \cup (T \cup U) = (S \cup T) \cup U$$
$$S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$$
$$S \cap S^C = \emptyset$$
$$S \cap \Omega = S$$

Recall: Set Theory

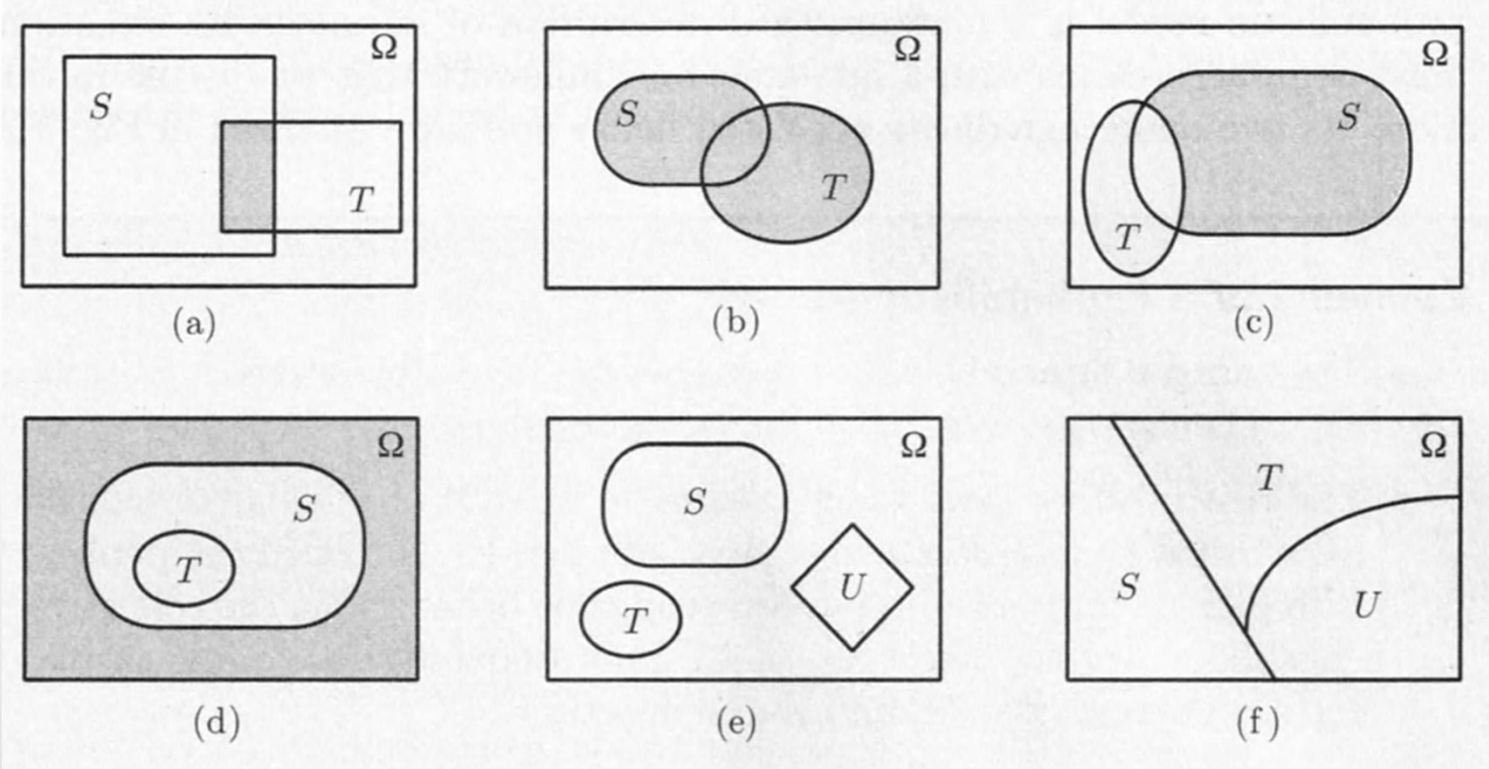
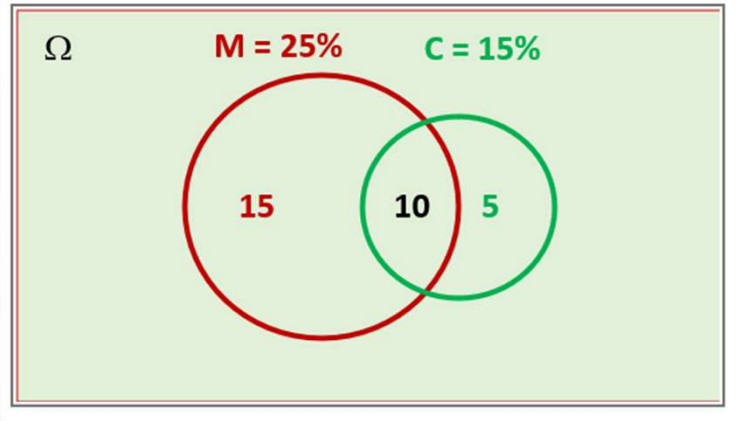


Figure 1.1: Examples of Venn diagrams. (a) The shaded region is $S \cap T$. (b) The shaded region is $S \cup T$. (c) The shaded region is $S \cap T^c$. (d) Here, $T \subset S$. The shaded region is the complement of S. (e) The sets S, T, and U are disjoint. (f) The sets S, T, and U form a partition of the set Ω.

Example:

- **Question**: In a certain college, 25% of the students failed mathematics, 15% of the students failed chemistry, and 10% of the students failed both mathematics and chemistry. A student is selected at random. **If he failed chemistry, what is the probability that he failed mathematics too?**



De Morgan's Laws

$$\left(\bigcup_n S_n\right)^c = \bigcap_n S_n^c$$

$$\left(\bigcap_n S_n\right)^c = \bigcup_n S_n^c$$

De Morgan's Laws Proof

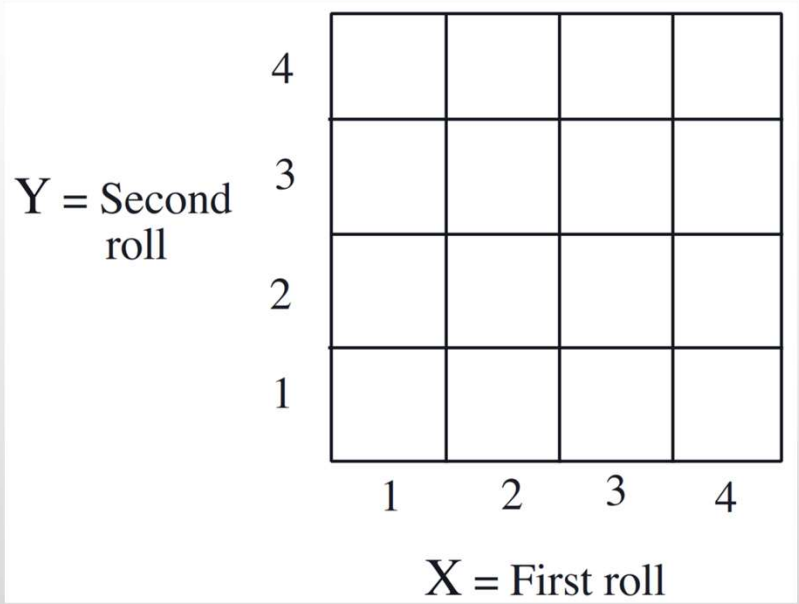
De Morgan's Laws Proof

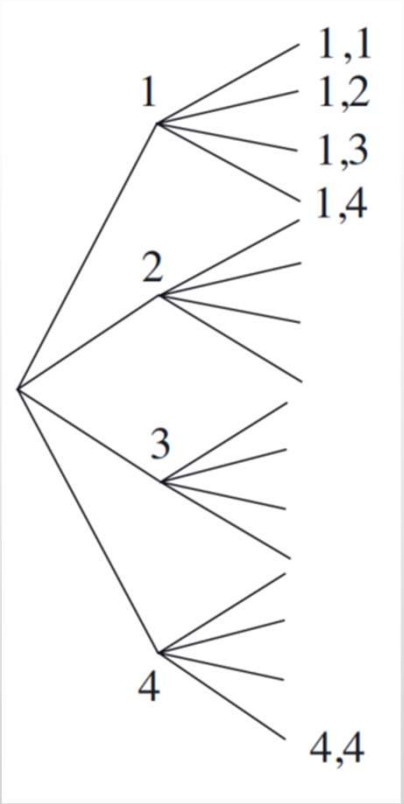
Sample Space Ω

- "List" (set) of possible outcomes
- List must be:
  - Mutually Exclusive
  - Collectively Exhaustive
- Rule of thumb: we want to be at the "right" granularity

Sample Space: Discrete Example

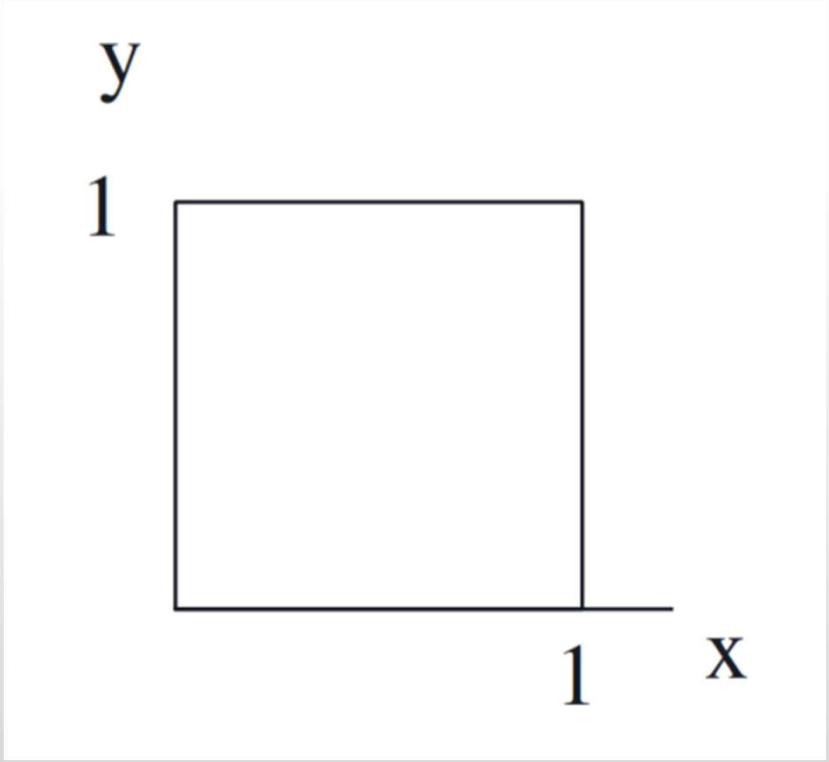
- Two Rolls of a tetrahedral die
  - Sample space vs. Sequential Description





Sample Space: Continuous Example

$$\Omega = \{(x, y) | 0 \leq x, y \leq 1\}$$



Probability Axioms

- **Event:** a subset of the sample space
- Probability is assigned to events

Axioms:

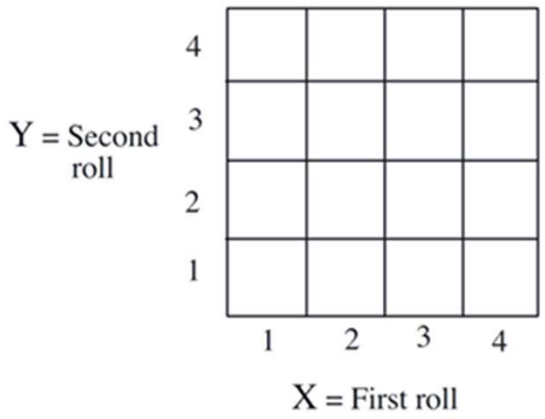
1. Nonnegativity: $\mathbf{P}(A) \geq 0$
2. Normalization: $\mathbf{P}(\Omega) = 1$
3. Additivity: If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

- $\mathbf{P}(\{s_1, s_2, \ldots, s_k\}) = \mathbf{P}(\{s_1\}) + \cdots + \mathbf{P}(\{s_k\})$
  $= \mathbf{P}(s_1) + \cdots + \mathbf{P}(s_k)$

Prove that the Probability for Event $A \leq 1$

For a coin toss, which of the following are valid Probability Laws?

Probability Law: Example with finite sample space



- Let every possible outcome have probability 1/16
  - $\mathbf{P}((X,Y)$ is $(1,1)$ or $(1,2)) =$
  - $\mathbf{P}(\{X = 1\}) =$
  - $\mathbf{P}(X + Y$ is odd$) =$
  - $\mathbf{P}(\min(X,Y) = 2) =$